Reverse-check review package — CHEIDI_gen

An independent, **blank-slate** agent instance reconstructed the physics below from the sanitized `.fr` alone (no paper, no metadata, no conversation history). The final verdict belongs to the human reviewer — sign off at the bottom.

item	value
model file	`CHEIDI/repair2/final.fr`
original model name	`CHEIDI_gen` (hidden from the agent)
paper	CHEIDI/text/1010.3251.txt
blindness scope	blank slate w.r.t. paper, authors, model name, comments and prose labels — NOT w.r.t. field/parameter symbol names, which are kept so the reviewer can map the reconstruction back

Verbatim Lagrangian terms (from the `.fr`)

These are the terms the reconstruction must account for, quoted unmodified. Check each against its reconstructed LaTeX form below.

`heidiVDefault` (=)

```
246.22
```

`heidiMhDefault` (=)

```
120.
```

`heidiCsDefault` (=)

```
100.
```

`heidiMbDefault` (=)

```
100.
```

`heidiG2Default` (=)

```
1.
```

`heidiCutoffDefault` (=)

```
1000.
```

`heidiNmodesDefault` (=)

```
10
```

`heidiModeRange` (=)

```
Range[heidiNmodesDefault]
```

`LHEIDI` (:=)

```
(LSM /. {muH -> 0, \[Lambda] -> 0}) /. H -> (Hh[n]*xi[n])
```

LHEIDIgg (:=)

```
Block[{fun, piece, tt, mm, hh}, tt[m_] := 4*MT^2/m^2; fun[tau_] := tau*(1 + (1 -
↪  tau)*If[NumericalValue[tau] > 1, ArcSin[Sqrt[1/tau]]^2, -1/4*(Log[(1 + Sqrt[1 + tau])/(1
↪  - Sqrt[1 - tau])] - I*Pi)^2]); mm[n_] := Symbol["mhh" <> ToString[n]]; hh[n_] :=
↪  Symbol["Hh" <> ToString[n]]; piece[n_] :=
↪  Sqrt[kggh]*gs^2/32/Pi^2/v*fun[tt[mm[n]]]*xi[n]*hh[n]*(del[G[mu, a], nu] - del[G[nu, a],
↪  mu])^2; Plus @@ (piece /@ heidiModeRange)]
```

LHEIDIggHeavyTop (:=)

```
Block[{piece, mm, hh}, mm[n_] := Symbol["mhh" <> ToString[n]]; hh[n_] := Symbol["Hh" <>
↪  ToString[n]]; piece[n_] := Sqrt[kggh]*gs^2/32/Pi^2/v*(2/3)*xi[n]*hh[n]*(del[G[mu, a],
↪  nu] - del[G[nu, a], mu])^2; Plus @@ (piece /@ heidiModeRange)]
```

LTot (:=)

```
LHEIDIggHeavyTop
```

Blank-slate reconstruction

Reconstruction of `sanitized.fr`

Lagrangian

Define the real scalar mode tower

$$S(x) \equiv \sum_{k=1}^{10} \xi_k \, Hh_k(x),$$

where `Hh[k]` is implemented as the class members `Hh1` through `Hh10`.

The new fields `Hh_k` have no declared SM gauge quantum numbers. Their covariant derivative is therefore just

$$D_\mu Hh_k = \partial_\mu Hh_k.$$

For SM fields appearing implicitly through `LSM`, the usual SM covariant derivative is understood:

$$D_\mu = \partial_\mu + ig_s T^a G^a_\mu + ig\,\frac{\tau^I}{2} W^I_\mu + ig' Y B_\mu,$$

with the relevant representation matrices omitted for gauge-singlet fields.

LHEIDI

The file defines

$$\texttt{LHEIDI} = \mathcal{L}_{\rm SM}\big|_{\mu_H\to 0,\ \lambda\to 0,\ H\to S}.$$

The new-field-dependent part is therefore the SM Higgs interaction structure with the physical Higgs field replaced by $S = \sum_k \xi_k Hh_k$, while the Higgs potential terms proportional to `muH` and `\[Lambda]` are removed.

Scalar kinetic term:

$$\boxed{\mathcal{L}_{\texttt{LHEIDI}} \supset \frac{1}{2}\,\partial_\mu S\,\partial^\mu S = \frac{1}{2}\sum_{k,\ell=1}^{10} \xi_k \xi_\ell\,\partial_\mu Hh_k\,\partial^\mu Hh_\ell}$$

Gauge-boson mass-interaction terms:

$$\boxed{\mathcal{L}_{\texttt{LHEIDI}} \supset \frac{2M_W^2}{v}\,S\,W^+_\mu W^{-\mu}}$$

$$\boxed{\mathcal{L}_{\texttt{LHEIDI}} \supset \frac{M_W^2}{v^2}\, S^2\, W_\mu^+ W^{-\mu}}$$

$$\boxed{\mathcal{L}_{\texttt{LHEIDI}} \supset \frac{M_Z^2}{v}\, S\, Z_\mu Z^\mu}$$

$$\boxed{\mathcal{L}_{\texttt{LHEIDI}} \supset \frac{M_Z^2}{2v^2}\, S^2\, Z_\mu Z^\mu}$$

Fermion Yukawa terms:

$$\boxed{\mathcal{L}_{\texttt{LHEIDI}} \supset -\sum_i \frac{m_{u_i}}{v}\, S\, \bar{u}_i u_i}$$

$$\boxed{\mathcal{L}_{\texttt{LHEIDI}} \supset -\sum_i \frac{m_{d_i}}{v}\, S\, \bar{d}_i d_i}$$

$$\boxed{\mathcal{L}_{\texttt{LHEIDI}} \supset -\sum_i \frac{m_{\ell_i}}{v}\, S\, \bar{\ell}_i \ell_i}$$

Equivalently, in chiral-projector notation,

$$\bar{f} f = \bar{f}_L f_R + \bar{f}_R f_L = \bar{f}\, P_R f + \bar{f}\, P_L f, \qquad P_{R,L} = \frac{1 \pm \gamma^5}{2}.$$

The file removes the SM Higgs potential through

$$\mu_H \to 0, \qquad \lambda \to 0,$$

so no S, S^2, S^3, or S^4 potential terms are generated by `LHEIDI`.

`LHEIDIgg`

The file also defines a finite-top-mass effective scalar-gluon-gluon interaction:

$$\boxed{\mathcal{L}_{\texttt{LHEIDIgg}} = \sum_{k=1}^{10} \frac{\sqrt{K_{ggh}}\, g_s^2}{32\pi^2 v}\, F(\tau_k)\, \xi_k\, Hh_k\, \left(\partial_\nu G^a_\mu - \partial_\mu G^a_\nu\right) \left(\partial^\nu G^{a\mu} - \partial^\mu G^{a\nu}\right)}$$

with

$$\tau_k = \frac{4M_T^2}{m_{Hh_k}^2},$$

and the loop form factor as encoded in the file:

$$F(\tau) = \tau \left[1 + (1-\tau)\, f(\tau)\right],$$

where

$$f(\tau) = \begin{cases} \arcsin^2\!\left(\sqrt{1/\tau}\right), & \text{if } \tau > 1, \\ -\dfrac{1}{4}\left[\log\!\left(\dfrac{1+\sqrt{1+\tau}}{1-\sqrt{1-\tau}}\right) - i\pi\right]^2, & \text{otherwise.} \end{cases}$$

Here G^a_μ is the gluon field. The `.fr` expression uses only the Abelianized derivative combination $\partial_\nu G^a_\mu - \partial_\mu G^a_\nu$, not the full non-Abelian field strength

$$G^a_{\mu\nu} = \partial_\mu G^a_\nu - \partial_\nu G^a_\mu + g_s f^{abc} G^b_\mu G^c_\nu.$$

`LHEIDIggHeavyTop`

The heavy-top limit replaces $F(\tau_k) \to 2/3$:

$$\boxed{\mathcal{L}_{\texttt{LHEIDIggHeavyTop}} = \sum_{k=1}^{10} \frac{\sqrt{K_{ggh}}\, g_s^2}{32\pi^2 v} \frac{2}{3}\, \xi_k \, Hh_k \, \left(\partial_\nu G^a_\mu - \partial_\mu G^a_\nu\right)\left(\partial^\nu G^{a\mu} - \partial^\mu G^{a\nu}\right)}$$

Equivalently,

$$\boxed{\mathcal{L}_{\texttt{LHEIDIggHeavyTop}} = \sum_{k=1}^{10} \frac{\sqrt{K_{ggh}}\, g_s^2}{48\pi^2 v}\, \xi_k \, Hh_k \, G^a_{\mu\nu,\mathrm{lin}} G^{a\mu\nu}_{\mathrm{lin}}}$$

with

$$G^a_{\mu\nu,\mathrm{lin}} = \partial_\mu G^a_\nu - \partial_\nu G^a_\mu.$$

`LTot`

The total Lagrangian exported by the file is

$$\boxed{\mathcal{L}_{\texttt{LTot}} = \mathcal{L}_{\texttt{LHEIDIggHeavyTop}}}$$

so the active `LTot` contains only the heavy-top effective scalar-gluon-gluon interaction, not the full `LHEIDI` expression.

Field Table

`.fr` symbol	Class members	Spin	SU(3) rep	SU(2) rep	U(1) charge / hypercharge	Self-conjugate	Mass symbol and value
`Hh`	`Hh1`	0	singlet	singlet	0	yes	`mhh1 = 120.000000`
`Hh`	`Hh2`	0	singlet	singlet	0	yes	`mhh2 = 141.421356`
`Hh`	`Hh3`	0	singlet	singlet	0	yes	`mhh3 = 223.606798`
`Hh`	`Hh4`	0	singlet	singlet	0	yes	`mhh4 = 316.227766`
`Hh`	`Hh5`	0	singlet	singlet	0	yes	`mhh5 = 412.310563`
`Hh`	`Hh6`	0	singlet	singlet	0	yes	`mhh6 = 509.901951`
`Hh`	`Hh7`	0	singlet	singlet	0	yes	`mhh7 = 608.276253`
`Hh`	`Hh8`	0	singlet	singlet	0	yes	`mhh8 = 707.106781`
`Hh`	`Hh9`	0	singlet	singlet	0	yes	`mhh9 = 806.225775`
`Hh`	`Hh10`	0	singlet	singlet	0	yes	`mhh10 = 905.538514`

The values shown are the built-in fallback values from

$$m_{Hh_1} = m_h, \qquad m_{Hh_k} = \sqrt{m_b^2 + (k-1)^2 M_c^2} \quad (k = 2, \dots, 10),$$

using

$$m_h = 120, \qquad m_b = 100, \qquad M_c = 100.$$

Widths are declared as

$$\Gamma_{Hh_k} = \texttt{whh}k,$$

with fallback numerical value 0 for all modes.

Parameters

Parameter	Default value	Appears in / multiplies	Physical meaning
`HeidiV`	`246.22`	Passed to `InitHeidi`; TeX v. The explicit Lagrangian terms use the SM symbol `v`.	Electroweak vacuum expectation value used to initialize the scalar-sector functions.
`HeidiMh`	`120.`	Sets the fallback value of `mhh1`.	Reference Higgs-like scalar mass parameter m_h.
`HeidiCs`	`100.`	Enters fallback masses $m_{Hh_k} = \sqrt{m_b^2 + (k-1)^2 M_c^2}$.	Mode-spacing / compactification-scale-like mass parameter M_c.
`HeidiMb`	`100.`	Enters fallback masses $m_{Hh_k} = \sqrt{m_b^2 + (k-1)^2 M_c^2}$.	Base scalar mass parameter m_b.
`HeidiG2`	`1.`	Passed to `InitHeidi`; not explicitly present in `LTot`.	Scalar-sector coupling-squared input, TeX g_5^2.
`HeidiCutoff`	`1000.`	Declared but not used directly in any displayed Lagrangian expression.	Cutoff parameter Λ.
`HeidiNmodes`	`10`	Defines `heidiModeRange = Range[10]`.	Number of scalar modes retained, $N_H = 10$.
`xi[k]`	fallback `xi[1]=1`, `xi[2..10]=0`	Multiplies each scalar mode in $S = \sum_k \xi_k Hh_k$ and in the effective $Hh_k gg$ interaction.	Mode wavefunction / mixing coefficient controlling the coupling strength of each scalar mode to SM Higgs-like interactions.
`kggh`	`1`	Appears as $\sqrt{K_{ggh}}$ multiplying `LHEIDIgg` and `LHEIDIggHeavyTop`.	Overall rescaling factor for the effective scalar-gluon-gluon coupling.

Physics Summary

The file encodes ten real gauge-singlet scalar modes `Hh1` through `Hh10` that couple to Standard-Model fields through Higgs-like mixing coefficients ξ_k. The only active total Lagrangian, `LTot`, keeps the heavy-top effective coupling of these scalars to two gluons, so it mediates gluon-fusion production $gg \to Hh_k$ and the crossed decay $Hh_k \to gg$.

Paper cross-check

(not run — provide `paper_tex_path` to enable the term-by-term comparison)

Suggested checks for the reviewer

1. Every verbatim `.fr` term above has a reconstructed LaTeX counterpart with the same field content, chirality and conjugation.
2. Kinetic terms: covariant derivative gauge content matches the field's representations; normalization is canonical.
3. Non-self-conjugate interaction terms appear together with their Hermitian conjugates.
4. Quantum numbers in the field table match the `.fr` declarations (and the paper, where the cross-check table flags disagreements).
5. Numeric masses/couplings are placeholders unless the paper pins them — treat values as demo inputs, not measurements.
6. Sanitizer scope: 9 prose labels were scrubbed; field/parameter symbol names were kept and may hint at the model's identity.

Physicist sign-off

- Reviewed by: ________________________________ Date: __________________
- Verdict: [ ] approve [ ] approve with corrections [ ] reject
- Notes: